Truthful Verified Credit: Making Truth Dominant for Exploring Agents

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Abstract

Learning agents that share claims are increasingly rewarded for being relied on, but credit for attention rewards being followed, not being honest. We show that such credit breeds hype, and that under congestion hype is a social dilemma: it benefits each agent yet ruins the group. We propose truthful verified credit (TVC), which rewards only verified value, adds a bonus for every checked claim, and fines audited false claims, and we derive the audit fine needed to outweigh the private outside gain of a lie. In a multi-agent bandit, TVC cuts exaggeration from over 90% of reports to below 1% at welfare comparable to popularity credit. In multi-robot pathfinding on POGEMA, it cuts the hype rate of self-interested learners from 99% to 0.2% and triples lifelong throughput, and lying collapses within a factor of two of the predicted fine threshold.

1. Introduction

In a review of FDA records for twelve antidepressants, 94% of the trials in the published literature were positive, against 51% in the FDA's own analysis, and the published effect sizes were 32% larger overall (Turner et al., 2008). The pattern is not confined to medicine: in a set of peer-reviewed social-science experiments, strong results were 60 percentage points more likely to be written up than null results (Franco et al., 2014). Selection also acts inside a study. Comparing trial protocols with their publications, 40–62% of trials had at least one primary outcome changed, introduced or omitted, and statistically significant outcomes were more likely to be fully reported (Dwan et al., 2008). This is publication bias and reporting bias, the file drawer problem of Rosenthal (1979), and it needs no fraud: small studies, flexible analyses and competing interests already make many published findings likely to be false (Ioannidis, 2005), and incentives that reward publication can select for weak methods without anyone cheating deliberately (Smaldino & McElreath, 2016). Authors respond to the credit they receive, and a literature that credits positive, attention-getting results teaches them to report only those.

Behind these figures lies a problem of rewarding exploration. Agents that search on behalf of a group bear the cost of exploring, but what they find can save everyone else the same work, so a group gains only when its members explore and then report truthfully what they find. Models of the division of cognitive labor show that the reward for discovery shapes which problems a community chooses to pursue (Kitcher, 1990; Strevens, 2003), and the same dependence appears when a planner decides what to disclose to agents whose own experiments create the information others rely on (Kremer et al., 2014). How to reward such exploration is a credit assignment problem, the problem of crediting each agent for its own contribution to a shared return (Foerster et al., 2018). A growing family of methods now rewards agents directly for their effect on others: social-influence rewards credit an agent for changing other agents' actions (Jaques et al., 2019), and learned incentive functions let agents reward one another to shape behavior (Yang et al., 2020). Whenever credit flows to whoever is relied on, it determines not only how much a group explores but whether its members report what they find honestly.

Theory explains why such failures persist. Conflicts of interest limit how much information survives strategic communication (Crawford & Sobel, 1982), receivers who take messages at face value are deceived by inflated language (Kartik et al., 2007), and a sender who controls what is revealed can steer a rational receiver's choice (Kamenica & Gentzkow, 2011). Citation counts can be inflated outright, as when editors pressure authors to cite their journal (Wilhite & Fong, 2012), and reward-maximizing learning agents with no built-in intent to deceive have been observed to distort their signals once distortion is profitable (Alon et al., 2023).

Intuitively, there are six common designs for the credit. However, each falls short.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

Credit from the shared return. Every agent receives the group's return, the default in cooperative multi-agent learning, where a shared reward does not say which agent contributed to it (Foerster et al., 2018). An explorer bears the full cost of exploring, while its own contribution to a return shared by many agents is hard to discern, so exploring and reporting are under-supplied. It also cannot tell a true report from an untruthful one, since both change the shared return only through what others later do with them.

Credit for keeping exploring. Explorers are credited for each exploration, whatever it yields. This sustains exploration, but it is blind to what is reported: a true report, an exaggerated one and silence all earn the same, so honesty has no advantage, and credit is spent on explorations whose findings never reach the group. Untruthful reports then harm the group that relies on them: they inflate apparent effects, as in the antidepressant trials above (Turner et al., 2008), and they add to avoidable waste in research (Chalmers & Glasziou, 2009).

Credit for only being followed (Popularity credit). Explorers are credited for attention, not accuracy. It encourages agents to explore more as they get rewarded more. A inflated claim attracts followers as readily as a real one, so exaggeration is profitable and a learner finds it. This is the same situation that celebrities may promote low-quality products without careful consideration, prioritizing their own financial gain over the well-being of consumers or reporting bias aforementioned.

Credit for only being accurate. Explorers are credited only for the verified gain of those who rely on a claim. This leaves a true report that nobody uses with nothing, which is the incentive behind the file drawer, and a community can stall in a state where no one reports because no one listens. Such credit also never checks an unused claim, so truth, exaggeration and silence all earn the same. Accuracy-based remedies from the literature do not close this gap: peer prediction makes honesty an equilibrium rather than a dominant strategy (Miller et al., 2005), and proper scoring rules require the outcome to be observed (Gneiting & Raftery, 2007).

A cost for lying alone. A fine charged when an audit finds a report false (Becker, 1968). It helps reduce lying rate of the agents. Nevertheless, it is not enough either: a lie that is followed still collects the followers' credit, so the fine has to outbid it, and a true report that nobody follows still earns nothing, so truth is no better than silence. Moreover, a lie can be profitable outside the credit channel, as when a hyped claim attracts followers whose attention benefits the claimant, so the penalty must outweigh a temptation the designer does not control. With smooth lying costs, (Kartik, 2009) shows that truthful reporting is never an equilibrium, and that inflation shrinks with cost intensity but persists.

Reputation without credit. Receivers stop following senders who were wrong before, and give nothing. This learns only from claims that were checked, and a claim that nobody follows is never checked, so unused lies leave a sender's reputation untouched. It also gives an honest explorer no reason to report: a new sender starts without trust and may not be followed, so a true report can go unused and earn nothing.

These failures set our goal. We ask for a credit rule under which reporting the truth is each agent's best action whatever the others do with the report, including when a lie or a withheld result carries a private gain the designer neither funds nor observes. Because receivers in learning systems, like readers of a literature, keep changing how they respond, the guarantee cannot rest on an equilibrium or on a model of receiver behavior; it must hold for every receiver rule, so truth must beat the response most favorable to a lie.

Our key idea is to reward only verified value and to make every claim face a check. We propose Truthful Verified Credit (TVC), which wraps any host credit scheme with three parts, each rewarding truth at least as much as a lie in every case. Verified gain credit rewards the realized improvement of those who rely on a report, capped, and only when the report passes verification. A checked score bonus rewards every report that is checked, whether by a follower or by an independent random audit, so a true report, including a negative one, earns credit even when nobody acts on it. An audit fine is charged when an audit finds a report false, and we derive the fine needed to outweigh a lie's outside gain.

We study TVC in two settings. In a multi-agent bandit, where agents search arms of unknown value on behalf of a group, popularity credit drives over 90% of reports to be exaggerated, while TVC brings it below one percent at comparable welfare. In a congestion setting built on a multi-agent pathfinding (MAPF) benchmark (Skrynnik et al., 2025), robots can post fake jams to clear their own routes: self-interested learners lie at 91–99% of opportunities without credit, TVC holds the rate near 0.2%, and throughput rises about threefold.

2. Related Work

Credit among learning agents. Credit assignment in cooperative multi-agent reinforcement learning asks how to credit each agent for its share of a shared return. Payoff design gives each agent a payoff that serves the group objective (Wolpert & Tumer, 2001), counterfactual baselines estimate an agent's marginal contribution (Foerster et al.,

2018), and value decomposition splits a team value into per-agent values (Sunehag et al., 2018; Rashid et al., 2018). These methods assume cooperative agents who report honestly. With mixed motives, as in sequential social dilemmas (Leibo et al., 2017), shaping the reward changes behavior: inequity aversion promotes cooperation (Hughes et al., 2018), social-influence rewards credit an agent for changing others' actions (Jaques et al., 2019), and gifting or learned incentives let agents reward one another (Lupu & Precup, 2020; Yang et al., 2020). These incentives reward being relied on, not being right, so an agent can raise its credit by exaggerating. This is an instance of reward hacking, where optimizing a proxy reward degrades the true objective (Amodei et al., 2016; Skalse et al., 2022). TVC is a rule for what to reward, and it can replace that credit.

Strategic communication. Conflicts of interest limit what cheap talk can convey (Crawford & Sobel, 1982). Receivers who take messages at face value are deceived by inflated language (Kartik et al., 2007), a sender who controls what is revealed can steer a rational receiver (Kamenica & Gentzkow, 2011), and with a smooth cost of lying, inflation falls but persists (Kartik, 2009). Learning agents with no intent to deceive distort their signals once distortion is profitable (Alon et al., 2023). These works analyze equilibria against a model of the receiver. We ask for dominance against every receiver rule.

Truthful elicitation and verification. Proper scoring rules need the outcome to be observed (Gneiting & Raftery, 2007), and peer prediction (Miller et al., 2005; Shnayder et al., 2016; Feng et al., 2022; Gao et al., 2019) makes honesty an equilibrium and not a dominant strategy. With costly verification, the optimal mechanism checks only some reports, and a cost of lying lowers how often (Ben-Porath et al., 2014), in line with deterrence by probabilistic enforcement (Becker, 1968). TVC audits a random sample, which makes truth dominant, and its follower gain is a VCG-style externality payment (Vickrey, 1961; Clarke, 1971; Groves, 1973) adapted to receivers who act by their own rule.

3. Problem Setting

We consider a multi-agent bandit game with private payoff. There are N agents and K arms. Arm k has a value $v_k > 0$, drawn i.i.d. from a distribution F with finite mean and fixed within an episode. The mean value is $\bar{v} = \frac{1}{K}\sum_k v_k$, and all credit amounts below are in units of $\bar{v}$. Play lasts T rounds. Agent i holds a set of known arms $\mathcal{K}_i^t \subseteq \{1, \dots, K\}$ with current best value $b_i^t = \max_{k \in \mathcal{K}_i^t} v_k$.

In each round, a free agent i chooses through its policy π_i to *explore*, pulling an unknown arm and adding it to $\mathcal{K}_i$, or to *exploit*, pulling its best known arm. Exploring costs $c \geq 0$. Agent i observes the payoff of its own pulls only, and its payoff u_i^t is the value of the pulled arm, minus c when exploring. On discovering arm k in round t, agent i observes $y_i^t = v_k e^{\epsilon_i^t}$ with $\epsilon_i^t \sim \mathcal{N}(0, \sigma^2)$, drawn i.i.d., is the observation noise: a random error that the explorer picks up when it discovers an arm. We write $\theta_i^t = v_k$ for the true value behind this observation. Pulling an arm yields its true value; only the discovery observation is noisy.

After discovering an arm, agent i may post a public claim $m_i^t \in \{\varnothing\} \cup \mathbb{R}_{>0}$, where $\varnothing$ is silence. Posting a claim cost $\xi \geq 0$. The claim is *truthful* if $m_i^t = y_i^t$. We denote $\kappa_i^t = m_i^t / y_i^t$ is inflation rate. Each free agent j that did not explore then chooses whom to follow, $d_j^t = \rho_j(h_j^t, m^t) \in \{0\} \cup \{1, \dots, N\} \setminus \{j\}$, where 0 means nobody, h_j^t is j's private history and $m^t = (m_1^t, \dots, m_N^t)$. Following i commits j to i's arm for L rounds. The *receiver rule* ρ_j is an arbitrary function: it may be a fixed heuristic, a Bayesian update, a learning policy or an adversary. We assume no rationality and no equilibrium among receivers. The follower's gain over its own best is

$$g_{ji}^t = L\left(\theta_i^t - b_j^t\right)^+. \tag{1}$$

A receiver cannot check a claim before it acts on it; it learns θ_i^t exactly only by following.

Reward. A sender maximises a mixed-motive reward in which it weighs the others' mean payoff $\bar{u}_{-i}^t$ by the *prosocial weight* $\lambda \in [0, 1]$, where $\lambda = 0$ is a selfish sender and $\lambda = 1$ a fully cooperative one,

$$r_i^t = (1 - \lambda)\, u_i^t + \lambda\, \bar{u}_{-i}^t - \xi\, \mathbb{1}[m_i^t \neq \varnothing] + C_i^t, \tag{2}$$

where $\bar{u}_{-i}^t = \frac{1}{N-1}\sum_{j \neq i} u_j^t$ is the mean payoff of the other agents.

Outside gain. A claim can also change the sender's reward outside the credit, because believers act on it. Relative to a truthful claim, a lie raises the liar's own payoff u_i^t by $G_1 \geq 0$, for example the citations a researcher gains, and lowers the others' mean payoff $\bar{u}_{-i}^t$ by $D_1 \geq 0$, for example the time they waste on a misleading approach, both per lie and in units of $\bar{v}$. Formally, $G_1 = \mathbb{E}[u_i^t \mid \text{lie}] - \mathbb{E}[u_i^t \mid \text{truthful claim}]$ and $D_1 = \mathbb{E}[\bar{u}_{-i}^t \mid \text{truthful claim}] - \mathbb{E}[\bar{u}_{-i}^t \mid \text{lie}]$, where the expectation is over how the receivers respond. A sender's payoff changes with a claim only through what others do, so $G_1 > 0$ needs a setting in which others' actions affect it. We take the largest values over receiver rules. The sender feels a share $1 - \lambda$ of its gain and a share λ of the damage, so a lie changes the part of its reward that does not come from the credit by the *outside gain*

$$\Gamma = (1 - \lambda)G_1 - \lambda D_1. \tag{3}$$

The designer neither observes nor provides Γ, and we assume $\Gamma \geq 0$. The posting cost ξ is borne by truth and lie

alike, so it does not enter Γ; it matters only when a claim is compared with silence.

4. Truthful Verified Credit

4.1. Verified externality credit

Popularity credit rewards a fake claim and a real one alike, because both attract followers. Let $\mathcal{F}_i^t = \{j : d_j^t = i\}$ be the *follow set* of sender i: the receivers j that follow i's claim in round t and so observe θ_i^t exactly. TVC instead credits the followers' realized gain Equation (1), and only for a claim that is verified:

$$\sum_{j \in \mathcal{F}_i^t} V_i^t \, \min\big(g_{ji}^t, \, \bar{B}\bar{v}\big). \tag{4}$$

Here $V_i^t = \mathbb{1}\big[\, |m_i^t/\theta_i^t - 1| \le \tau \,\big]$ is the verification gate: it equals 1 when the claim lies within a tolerance $\tau \ge 0$ of the true value θ_i^t, and 0 otherwise. The tolerance τ is set above the honest noise level ϵ, so a truthful claim passes the gate and any claim outside the tolerance is blocked. Every $j \in \mathcal{F}_i^t$ learns θ_i^t exactly on committing, so a followed claim ($\mathcal{F}_i^t \neq \emptyset$) is checked at no cost to the designer.

The gain g_{ji}^t is the externality of i's report on j relative to j staying put, as in a VCG mechanism (Vickrey, 1961; Clarke, 1971; Groves, 1973). A lie that is followed fails the gate V_i^t and earns no gain credit, so attracting followers with an inflated claim no longer brings a gain. The cap $\bar{B}\bar{v}$, with $\bar{B} > 0$ a design parameter in units of the mean value $\bar{v}$, limits what one follower can add, so a single lucky find cannot dominate the sender's return. Each receiver follows at most one claim, so the cap also bounds the cost: the gain term adds at most $N\bar{B}\bar{v}$ per round.

4.2. Checked score bonus

Gain credit alone leaves silence as a best response: a true claim that nobody follows earns nothing, exactly like staying silent. TVC therefore gives every checked claim a bonus that peaks at the truth,

$$\chi_i^t \, \eta \, \bar{v} \, S\big(m_i^t, \theta_i^t\big), \tag{5}$$

where $\eta > 0$ is the bonus weight and the score is a Gaussian in the log error with width $w > 0$,

$$S(m, \theta) = \exp\Big(- \frac{(\ln m - \ln \theta)^2}{2w^2} \Big). \tag{6}$$

A claim is *checked* when it is audited or followed,

$$\chi_i^t = \mathbb{1}[m_i^t \neq \varnothing] \, \max\big(a_i^t, f_i^t\big), \qquad f_i^t = \mathbb{1}\big[\mathcal{F}_i^t \neq \emptyset\big], \tag{7}$$

where f_i^t indicates that someone followed i, and a_i^t indicates a random audit of the claim, drawn independently of the claim and of the receivers,

$$a_i^t \sim \text{Bernoulli}(p), \qquad p \in (0, 1]. \tag{8}$$

A claim is thus checked either by a follower or by a random audit, and the audit is what reaches claims nobody follows. Because the audit is independent of the claim and of the receivers, a true claim earns at least $p\,\eta\bar{v}$ in expectation however receivers behave, and this is strictly more than silence, which earns 0. This holds for a bad find as well, so reporting a low-value find is no longer a loss. The score is 1 at the truth and below $\bar{s} = \sup_{|m/\theta - 1| > \tau} S(m, \theta) < 1$ for every claim that fails the tolerance, so inflation cannot raise the bonus.

4.3. Audited lying cost

The first two parts leave a lie with a small payoff, the bonus up to $\eta\bar{s}\bar{v}$, and say nothing about an outside gain Γ, the net private benefit a lie brings its sender outside the credit channel, for example a false report that steers others away from something the sender wants for itself. TVC therefore fines every audited claim that fails verification, with a fine $\varphi \ge 0$ in units of $\bar{v}$:

$$a_i^t \, (1 - V_i^t) \, \mathbb{1}[m_i^t \neq \varnothing] \, \varphi \, \bar{v}. \tag{9}$$

The fine is charged only on audits. An audit has probability p, independent of the claim and the receivers, so a lie costs $p\varphi\bar{v}$ in expectation whatever the receivers do, and the fine can be sized against Γ without a model of the receivers (Equation (11)). Raising φ lets the designer lower the audit rate p, which is the price of verification. We do not assume limited liability, so the net credit may be negative.

Together, the credit to sender i in round t is

$$\begin{aligned} C_i^t = \; & \textstyle\sum_{j \in \mathcal{F}_i^t} V_i^t \min\big(g_{ji}^t, \bar{B}\bar{v}\big) \\ & + \chi_i^t \, \eta \, \bar{v} \, S\big(m_i^t, \theta_i^t\big) \\ & - a_i^t \big(1 - V_i^t\big) \mathbb{1}\big[m_i^t \neq \varnothing\big] \, \varphi \, \bar{v}. \end{aligned} \tag{10}$$

The algorithm for learning with TVC per episode is shown in Algorithm 1, and the credit of one claim in Algorithm 2.

4.4. When TVC is useful

TVC is aimed at settings where four conditions hold together.

1. *Credit follows reliance.* Agents are credited for being followed or used, which rewards exaggeration and leaves a true report that nobody uses uncredited.
2. *Lying is profitable outside the credit.* Agents are not fully prosocial ($\lambda < 1$), so a lie can bring a private gain that no reliance credit removes by itself.

Algorithm 1 Learning with TVC (per episode)

Each sender decides to post a claim m_i^t or stay silent; draw an audit $a_i^t \sim \text{Bernoulli}(p)$ per claim
Receivers act on the claims by their own rule; record who relied on which claim
Mark a claim checked if it was audited or its outcome was observed by someone who relied on it
for each claim **do**
  Compute C_i^t by Algorithm 2
  Add C_i^t to the return of the decision that posted the claim
end for
Update each sender's policy with its usual learning rule

Algorithm 2 TVC credit of one sender in one round (the credit step of Algorithm 1)

Require: claim m_i^t or silence $\varnothing$; follow set $\mathcal{F}_i^t$ and gains g_{ji}^t; audit draw a_i^t; true value θ_i^t, revealed by a check; parameters $\tau, \bar{B}, \eta, w, \varphi$
Ensure: credit C_i^t
if $m_i^t = \varnothing$ **then**
  return 0
end if
$\chi_i^t \leftarrow \max\left(a_i^t,\ \mathbb{1}[\mathcal{F}_i^t \neq \emptyset]\right)$ {checked by an audit or a follower}
if $\chi_i^t = 0$ **then**
  return 0 {an unchecked claim earns nothing}
end if
$V_i^t \leftarrow \mathbb{1}\left[\,|m_i^t/\theta_i^t - 1| \le \tau\,\right]$ {verification gate}
$C_i^t \leftarrow \eta\,\bar{v}\,S\left(m_i^t, \theta_i^t\right)$ {score bonus}
if $V_i^t = 1$ **then**
  $C_i^t \leftarrow C_i^t + \sum_{j \in \mathcal{F}_i^t} \min\left(g_{ji}^t, \bar{B}\bar{v}\right)$ {verified gain credit}
else if $a_i^t = 1$ **then**
  $C_i^t \leftarrow C_i^t - \varphi\,\bar{v}$ {fine for an audited false claim}
end if
return C_i^t

3. *Lying harms others.* A false report misleads those who act on it and wastes their effort, and the group loses once several agents lie. Where receivers can decode exaggeration at little cost, hype costs little, and TVC buys honest reports but not extra welfare.

4. *Claims can be audited at a cost.* Receivers cannot verify a claim before acting, but the designer can check a random sample at rate p.

One of our testbeds meets all four (Section 5). Other candidates, which we do not evaluate, are crowd-sourced traffic or robot-fleet reports, teams of language-model agents that build on one another's findings, and scientific credit through citations with sampled replication as the audit. When a condition fails, simpler designs may suffice: a shared return can work for nearly altruistic agents, and without any audit, peer prediction is the relevant regime.

eter recipe: η in units of an value $\bar{v}$; the audit bud- $= 2\varphi_{1\text{o}}$ from on (11). Cost: dit per $1/p$

5. Experiments

We study TVC in two settings. In a multi-agent bandit, where agents search arms of unknown value on behalf of a group. In a congestion setting built on a multi-agent pathfinding (MAPF) benchmark (Skrynnik et al., 2025), robots can post fake jams to clear their own routes.

5.1. Experiment 1: Multi-agent Bandit

Setup. $N = 10$ agents search $K = 50$ arms whose values are drawn i.i.d. from a lognormal distribution, in episodes of $T = 100$ rounds, and each agent starts knowing one random arm. In each round a free agent either explores a random arm, at a cost of $0.5\bar{v}$, or exploits its best known arm. An explorer may post a claim about its find: silence, or the true value scaled by $\times 1$ (truthful), $\times 1.1$, $\times 1.5$ or $\times 2$. Free agents that did not explore follow the claim that beats their own best and stay committed for $L = 3$ rounds. We use two kinds of receiver: naive receivers take a claim at face value, and calibrated receivers learn, for each sender, how far its claims exceed the truth, from the claims they have followed, with learning rate 0.3. Explore and report decisions are learned by tabular REINFORCE (learning rate 0.1, discount 0.9, a running baseline per state), with the agent's own best and the value of its find as states.

Credit designs. We compare no credit, popularity credit ($2\bar{v}$ per follower), accuracy credit (the follower's signed gain), TVC ($\tau = 0.05$, $\bar{B} = 8$, $\eta = 2$, $p = 0.2$, $\varphi = 8$, $w = 0.05$), a shared return (every agent receives the group's mean payoff), credit for keeping exploring (a flat $3\bar{v}$ per exploration, matched to TVC's spend), a lying cost alone (audited false claims fined φ, nothing else given) and popularity credit with that fine. Reputation alone is no credit with calibrated receivers.

Metrics. Inflation (share of reports that are exaggerated), report rate (reports per exploration), misled follows (follows that lead to a worse arm), welfare (own payoff per round, in units of $\bar{v}$) and exploration rate. We report the mean over the last 100 of 1000 episodes with 95% confidence intervals over 32 independent runs.

Results. Table 1 compares the credit designs. Credit for being relied on breeds hype. With naive receivers, popularity credit raises inflation from 75.1% (no credit) to 94.0% and the share of misled follows from 25.5% to 43.5%. Accuracy

credit does not remove it (68.9% with naive and 85.1% with calibrated receivers). TVC brings inflation down to 0.3% with naive and 0.4% with calibrated receivers, raises the report rate to 99.6%, and cuts misled follows to 0.2%. A lying cost alone also removes most inflation (3.4%), but half of the explorers choose to silent (report rate 50.0%), and it has the lowest welfare (3.18 against 3.49 with no credit). Popularity credit with the same fine still has 38.4% inflation. In welfare, TVC (4.03) matches popularity credit (4.04) and exceeds no credit (3.49). The highest welfare comes from keeping exploring (4.28), which raises the exploration rate to 15.0% against 3.0% for TVC but leaves inflation at 74.7%. The shared return also raises exploration (23.3%) but reaches the same welfare as TVC (4.06) with 71.9% inflation. TVC therefore makes reports honest at no cost in welfare, but it does not by itself increase exploration. Calibrated receivers lower the cost of hype, as the conditions in Section 4.4 predict. With no credit they cut misled follows from 25.5% to 7.6%, and under popularity credit to 0.9%, so popularity credit reaches the welfare of TVC (4.02 against 3.96, within the intervals) at 92.2% inflation.

5.2. Experiment 2: MAPF with fake jams

Setup. We use POGEMA's lifelong pathfinding task (Skrynnik et al., 2025) with soft collisions on a 3×3 layout of 5×5 rooms joined by 1-cell doorways, with 12 robots, observation radius 3 and 256 steps. Robots move by greedy descent on a distance map, so the only learned decision is whether to post a claim that a doorway they can see is jammed. We define the ground truth that a doorway is jammed when at least 3 robots are within 2 walking steps of it. Receivers that believe a claim avoid that doorway for 20 steps, and a claim expires after 20 steps. A claim is checked by a robot that passes the doorway, or by an audit. Naive receivers take claims at face value, and sceptical receivers learn per-sender trust from the checks they see (learning rate 0.3).

Mixed motive. A robot's reward is $(1 - \lambda)$ times its own deliveries plus λ times the others' mean deliveries, minus a cost per claim posted, plus credit. A lie, a fake jam on its own route, gains the liar when believers leave that route and costs the others their detours. The per-believer gain, harm and benefit that enter the learning signal are calibrated from short runs with hand-set policies, and the temptation is therefore a modelling choice, not an emergent property. Claim policies are learned by tabular REINFORCE.

Credit designs. No credit, popularity credit (1 per believer), accuracy credit (the steps a believer saves, negative for a fake jam), TVC ($\tau = 0.05$, $\bar{B} = 8$, $\eta = 2$, $p = 0.2$, $\varphi = 15$), reputation (no credit, sceptical receivers) and a shared return ($\lambda = (N - 1)/N$).

Metrics. We report the lie rate (the share of opportunities on which a robot posts a fake jam on its own route), the useful-report rate (the share of real jams elsewhere that it reports), POGEMA's throughput (deliveries per step, summed over the 12 robots) and the credit given, as the mean over the last 50 of 150 episodes, with 95% confidence intervals over 10 seeds.

Questions. We ask questions. Does lying harm the group, so that honest reports are worth rewarding? Does TVC remove lying and raise throughput compared with the other credit designs? Does it do so at every prosocial weight?.

Lying harms the group. We first check that lying is costly for the group, because only then is it worth rewarding honest reports. We fix the claim policies instead of learning them. The first k of the 12 robots post a fake jam whenever a doorway on their own route is free, the others report real jams, and no credit is given. We run 10 batches of 20 episodes for each k, with 95% confidence intervals over batches. With naive receivers, group deliveries per robot per episode fall from 5.8 with no liars to 1.9 with 12 (Figure 1), and the share of detours that were for fake jams rises from 0.31 at $k = 1$ to 0.95 at $k = 12$. One liar gains 3.5 deliveries over its honest peers while the others lose 0.5, so a single lie mostly moves deliveries from the others to the liar, and the group loses clearly once several robots lie. Sceptical receivers, which learn a trust for each sender from the checks they see, soften this: the group falls to 3.7 at $k = 12$, and a liar's advantage at $k = 1$ falls from 3.5 to 1.6. A lying robot still delivers more than an honest one at every $k < 12$. Each episode starts with full trust, so this understates what trust learned over many episodes would do.

TVC removes lying and raises throughput. Table 2 gives the lifelong comparison at $\lambda = 0$. Without credit, selfish robots post a fake jam on 99.2% of their opportunities and report 44.7% of the real jams elsewhere. Popularity credit leaves the lie rate at 99.9% and raises useful reports to 94.6%, but throughput falls from 0.080 to 0.037, as in the hype sweep, where throughput falls as more robots lie (Figure 1). TVC cuts the lie rate to 0.2%, raises useful reports to 98.8%, and raises throughput to 0.266, 3.3 times that of no credit and 7.2 times that of popularity credit. The other designs fall between these. Reputation alone, that is sceptical receivers with no credit, keeps the lie rate at 80.1% and reaches only about 1.5 times the throughput of no credit. Accuracy credit removes lying (0.1%) by silencing the robots, with 1.2% useful reports, and its throughput is below that of no credit (about 0.75 times). A shared return with $\lambda = (N - 1)/N$ is the closest competitor: it cuts lying to 6.3% and reaches about 3.1 times the throughput of no credit, but it needs every robot to weigh the others' deliveries as much as its own, which TVC does not.

Table 1. Multi-agent bandit: credit designs compared (mean ± 95% CI over 32 runs, last 100 of 1000 episodes). Inflation is the percentage of reports that are exaggerated, report is the percentage of explorations that are reported, misled is the percentage of follows that lead to a worse arm, and explore is the percentage of agent-rounds spent exploring. Welfare is the mean own payoff per agent per round, in units of the mean arm value. With calibrated receivers, *None* is reputation only. *Keeping exploring* gives a flat $3\bar{v}$ per exploration, *Lying cost alone* fines audited false claims and gives nothing else, and *Shared return* gives every agent the group mean payoff.

Credit design	Inflation (%)	Report (%)	Misled (%)	Welfare ($\bar{v}$)	Explore (%)
Naive receivers					
None	75.1 ± 0.4	80.2 ± 0.3	25.5 ± 1.0	3.486 ± 0.104	2.7 ± 0.2
Popularity	94.0 ± 0.5	94.8 ± 0.4	43.5 ± 1.1	4.036 ± 0.054	4.5 ± 0.2
Accuracy	68.9 ± 2.4	89.8 ± 0.4	13.7 ± 1.2	3.867 ± 0.095	2.9 ± 0.2
Shared return	71.9 ± 5.4	80.1 ± 4.7	32.7 ± 4.8	4.055 ± 0.145	$\mathbf{23.3 \pm 3.0}$
Keeping exploring	74.7 ± 1.1	79.1 ± 1.2	41.3 ± 1.8	$\mathbf{4.280 \pm 0.083}$	15.0 ± 2.5
Lying cost alone	3.4 ± 0.7	50.0 ± 2.9	1.4 ± 0.3	3.177 ± 0.114	2.4 ± 0.2
Popularity + fine	38.4 ± 1.6	85.0 ± 2.4	34.4 ± 1.6	3.859 ± 0.070	3.4 ± 0.2
TVC	$\mathbf{0.3 \pm 0.1}$	$\mathbf{99.6 \pm 0.1}$	$\mathbf{0.2 \pm 0.1}$	4.028 ± 0.079	3.0 ± 0.2
Calibrated receivers (per-sender decoding)					
None	75.1 ± 0.5	80.1 ± 0.4	7.6 ± 0.5	3.528 ± 0.126	2.5 ± 0.3
Popularity	92.2 ± 0.5	93.4 ± 0.5	0.9 ± 0.4	4.023 ± 0.078	3.2 ± 0.2
Accuracy	85.1 ± 1.3	90.6 ± 0.4	3.3 ± 0.5	3.993 ± 0.097	3.1 ± 0.3
Shared return	68.0 ± 6.8	83.6 ± 3.4	3.3 ± 0.7	4.118 ± 0.159	$\mathbf{23.7 \pm 3.6}$
Keeping exploring	74.8 ± 1.2	79.2 ± 1.1	9.1 ± 0.6	$\mathbf{4.358 \pm 0.079}$	15.1 ± 2.6
Lying cost alone	3.5 ± 0.7	50.4 ± 3.6	1.3 ± 0.2	3.045 ± 0.128	2.2 ± 0.3
Popularity + fine	33.4 ± 2.3	85.4 ± 2.5	2.5 ± 0.4	3.877 ± 0.094	2.5 ± 0.2
TVC	$\mathbf{0.4 \pm 0.1}$	$\mathbf{99.6 \pm 0.2}$	$\mathbf{0.2 \pm 0.1}$	3.963 ± 0.081	2.9 ± 0.2

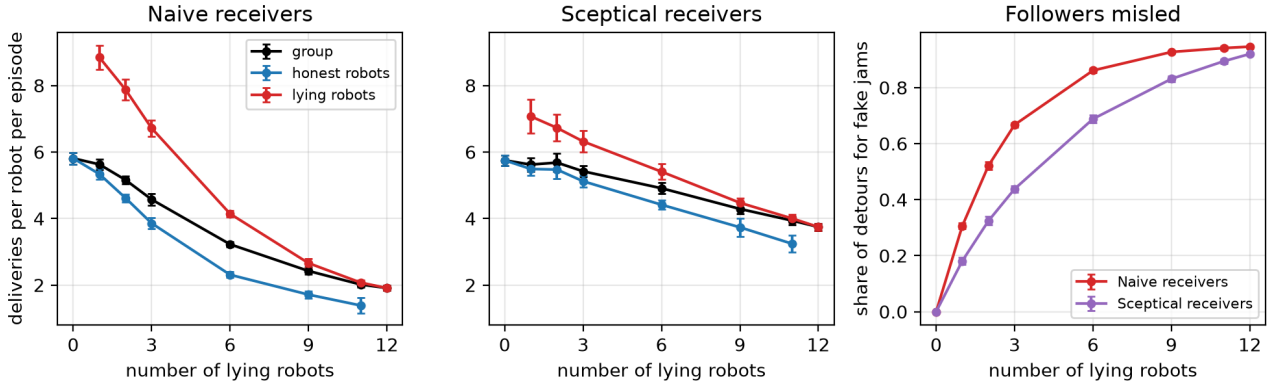


Figure 1. Lying harms the group in the congestion setting. The first k of 12 robots post fake jams and the rest report real jams, with fixed policies and no credit (10 batches of 20 episodes, 95% CIs). Left and middle: deliveries per robot per episode for the group, the honest robots and the lying robots, with naive and with sceptical receivers. Right: share of detours that were for fake jams.

Table 2. POGEMA lifelong MAPF ($\lambda = 0$, 10 seeds, mean ± 95% CI). Lie rate is the percentage of opportunities on which a robot posts a fake jam on its own route, useful is the percentage of real jams elsewhere that it reports, and throughput is POGEMA's `avg_throughput`, the deliveries per step summed over the 12 robots.

Rule	Lie rate (%)	Useful (%)	Throughput
None	99.2 ± 0.0	44.7 ± 1.5	0.080 ± 0.002
Popularity	99.9 ± 0.0	94.6 ± 0.8	0.037 ± 0.001
TVC	$\mathbf{0.2 \pm 0.1}$	$\mathbf{98.8 \pm 0.7}$	$\mathbf{0.266 \pm 0.007}$

TVC holds at every prosocial weight. Across the prosocial weight λ (Figure 2), popularity credit keeps the lie rate at 99.8–99.9% and throughput at 0.003, and TVC holds the lie rate at 0.2% or less at every λ, with throughput 0.022–0.023. With no credit, lying is 99.2% at $\lambda = 0$ and still 91.9% at $\lambda = 0.75$, and falls to 2.5% only at $\lambda = 1$, where throughput rises to 0.021. This agrees with the calibrated threshold, $\lambda \approx 0.93$, below which a lie is profitable without credit.

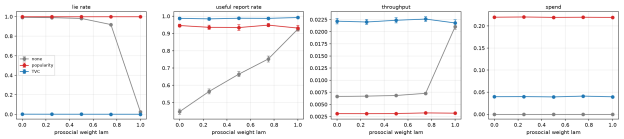


Figure 2. Learned behaviour by credit rule and prosocial weight λ (10 seeds, 95% CIs).

6. Conclusion

Conclusion. Credit that rewards agents for being relied on rewards attention, not accuracy, and learning agents respond by exaggerating. We proposed truthful verified credit (TVC), which gives credit only for verified value, adds a bonus for every checked claim, and fines audited false claims. The fine needed to outweigh a lie's outside gain grows as $1/p$ in the audit rate p. Across both experiments, TVC nearly eliminates exaggeration and keeps agents reporting. Its welfare is comparable to popularity credit when receivers can decode exaggeration cheaply, and well above both baselines when receivers act on claims. Lying stays low at every prosocial weight and collapses close to the theoretical fine threshold. TVC is called for when credit follows reliance, lying is profitable outside the credit, lying harms others, and a sample of claims can be audited. When agents weigh the others' payoffs almost as much as their own, a shared return alone comes close.

Limitations. TVC elicits truth, not usefulness. In the bandit its welfare matches popularity credit but does not exceed it, because calibrated receivers decode exaggeration cheaply, and designs that raise exploration reach higher welfare without making reports honest. The fine rule is derived for a sender that maximises expected reward in a single round. It also assumes that checks reveal the truth exactly, that audits are independent, and that receivers never follow a claim that cannot help them. Our testbeds are simulated, and the outside gain of a lie is calibrated from short runs, not an emergent property. The bonus accounts for most of the credit given in the congestion setting, yet removing it changes little there. Future work includes noisy verification, the cost of audits, and fleets with real data.

Impact Statement

This paper studies incentive design for learning agents that share information. Mechanisms that reward truthful, verified reports could make multi-agent systems more reliable and could reduce manipulation of shared information channels. The same tools could be misused to design audits that are unfair to some agents, and our design assumes accurate audits; deployments should account for audit errors. Beyond these points, we see no societal consequences that must be specifically highlighted here.

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Table 3. Reward r of a sender i under TVC in one round, based on Equation (2), in units of $\bar{v}$. Each row is what happens to the claim: whether some receiver follows it and whether the random audit checks it. Each column is what the sender does: report truthfully ($m = \theta$), report untruthfully ($m \neq \theta$, exaggerated or understated) or stay silent ($m = \varnothing$). Each cell is Equation (2) evaluated with the credit of Equation (10) for the case in its row and column, with $S = S(m, \theta)$. Every claim bears the posting cost ξ. An untruthful report never passes the verification gate, so it earns no gain credit, and it gains the outside gain Γ only when it is followed, because only then do believers act on it. We take the outside effect of a truthful claim as zero, since a positive effect only favours truth. Silence posts no claim, so it bears no cost and earns 0.

Case	Truth	Untruthful	Silence
Followed, audited	$\min(g, \bar{B}) + \eta - \xi$	$\Gamma + \eta S - \varphi - \xi$	0
Followed, not audited	$\min(g, \bar{B}) + \eta - \xi$	$\Gamma + \eta S - \xi$	0
Not followed, audited	$\eta - \xi$	$\eta S - \varphi - \xi$	0
Not followed, not audited	$-\xi$	$-\xi$	0

A. Analysis of Truthful Reporting in MAB with private payoff

Proposition A.1 (Truthful reporting is interim strictly dominant in one round)**.** *In one round, truthful reporting is a interim strictly dominant strategy for the sender Equation* (2) *if* $p\,(\eta + \varphi) > \eta\,\bar{s} + \Gamma$ *and* $p\eta > \xi$.

Proof. Let us reconsider the problem introduced in Section 3. We relax the problem into a single round with one sender. Thus, for brevity, we will drop the "t" notation. In this case, a *sender* i found an arm, and choose to make an report $m \in \{\varnothing\} \cup \mathbb{R}_{>0}$. A report m is truthful if $m = \theta$ and untruthful otherwise, whether exaggerated ($m > \theta$) or understated ($m < \theta$), and $m = \varnothing$ is silence. Everyone else, receivers and other senders, is collapsed into an arbitrary *opponent profile*, denoted σ_{-i}: any map from the posted claim to a distribution over their actions.

We assume the follows.

A1 *Separability.* Given the action profile, u_i does not depend on m_i. Claims affect i's reward only through C_i and through receivers' future behaviour.

A2 *Unforgeable verification.* On checked claims the designer learns θ_i exactly, and followers cannot misreport it.

A3 *Independent audits.* a_i is independent of (m, d, h).

A4 *Untruthful reports are detected.* Every untruthful report, whether exaggerated or understated, misses the tolerance: $m \neq \theta \Rightarrow |m/\theta - 1| > \tau$.

A5 *Truthful reports do not hurt the sender.* A truthful report does not lower the sender's weighted payoff $(1 - \lambda)u_i + \lambda\bar{u}_{-i}$ relative to silence. This holds when receivers never follow a truthful claim at a loss to themselves and following does not change the sender's own payoff.

Table 3 shows the reward r_i of a sender under TVC in one round.

Fix any opponent profile σ_{-i}, and write $O(m)$ for the expected value of $(1 - \lambda)u_i + \lambda\bar{u}_{-i}$ when the sender reports m. By separability (A1), $O(m)$ depends on m only through how the opponents respond, so $O(m) \leq O(\theta) + \Gamma$ for every untruthful report m, and $O(\theta) \geq O(\varnothing)$ by A5. The audit is independent of the claim and the receivers (A3), so a non-silent claim is checked with probability at least p, and a report that fails the gate is fined with probability exactly p. By *interim strict dominance* we mean that, for every true value θ and every opponent profile σ_{-i}, the expected reward of truthful reporting exceeds that of every other report. The sender knows its own find θ when it reports, but not yet the audit draw or the receivers' response, and the expectation is over those two. It does not require truth to be best in every realisation, and it is not (silence is better in the last row of Table 3). Write $\bar{r}(m)$ for the expected reward of report m.

A truthful claim passes the gate and its gain credit is nonnegative. It earns the bonus η whenever it is checked, which happens with probability at least p. So, whatever the receivers do,

$$\bar{r}(\theta) \;\geq\; O(\theta) - \xi + p\eta.$$

By A4 an untruthful report fails the gate, so it earns no gain credit and is fined φ when audited. Its bonus is at most $\eta\bar{s}$, because its score is at most $\bar{s}$ and it is checked at most always. So, whatever the receivers do,

$$\bar{r}(m) \;\le\; O(\theta) + \Gamma - \xi + \eta\bar{s} - p\varphi.$$

Silence bears no cost and cannot be followed or audited, so $\bar{r}(\varnothing) = O(\varnothing) \le O(\theta)$.

Subtracting, for every value θ, every untruthful report $m \neq \theta$ and every σ_{-i},

$$\bar{r}(\theta) - \bar{r}(m) \;\ge\; p(\eta+\varphi) - \eta\bar{s} - \Gamma \;>\; 0 \qquad \text{and} \qquad \bar{r}(\theta) - \bar{r}(\varnothing) \;\ge\; p\eta - \xi \;>\; 0,$$

both by the conditions of Proposition A.1. Neither bound depends on m or on σ_{-i}, so truthful reporting is interim strictly dominant. □

B. Sizing the Fine

We size the fine against the outside gain Γ of a lie (Equation (3)) and the posting cost ξ in the reward Equation (2).

By the analysis above, truth earns at least $p\eta\bar{v}$ from the credit, and a lie earns at most $(\eta\bar{s} - p\varphi)\bar{v}$ from the credit plus Γ outside it. Both bear the posting cost ξ, which cancels. Truth stays strictly better than a lie when $p\eta > \eta\bar{s} - p\varphi + \Gamma$, which gives

$$\varphi > \varphi_{\text{lo}} = \frac{\eta\,\bar{s} + \Gamma}{p} - \eta. \tag{11}$$

A larger outside gain needs a larger fine, and fewer audits (a smaller p) need a larger fine too, in proportion to $1/p$. If lies score almost nothing ($\bar{s} \approx 0$), the condition reads $\varphi > \Gamma/p - \eta$. Against silence, a truthful claim earns at least $p\eta\bar{v}$ of credit and bears the posting cost, so truth beats silence when $p\eta\bar{v} > \xi$, assuming a truthful claim has no negative outside effect. When the sender has no truthful claim to make, as at a free doorway in the congestion setting, its alternative to a lie is silence, which bears no posting cost, so ξ does not cancel. A lie then beats silence unless $\varphi > (\eta\bar{s} + \Gamma - \xi)/p$, which is $(\Gamma - \xi)/p$ when $\bar{s} \approx 0$. In the congestion experiment, φ_{lo} denotes this threshold.

C. Notation

Table 4 lists the symbols used in the paper. Amounts of credit are in units of the mean arm value $\bar{v}$.

D. Experimental Details

D.1. Hyperparameters

Tables 5 and 6 list every setting of the two experiments. Credit amounts are in units of the mean arm value $\bar{v}$ in the bandit, and in deliveries in the congestion setting.

D.2. Calibrating the Temptation

The congestion setting needs the per-believer gain, harm and benefit of a claim, which enter the sender's learning signal (Equation (2)). We measure them before training, once for each layout, with no credit and with hand-set claim policies on common random seeds, 30 episodes each. In every policy only robot 0 is the sender and the other robots stay silent, except in the last. The *silent* policy never posts. The *reporter* posts true jams elsewhere, and the *liar* posts fake jams on its own route. In *all-liars* every robot posts fake jams on its own route. Writing $T_\bullet$ for the mean deliveries per robot under policy $\bullet$ and $b_\bullet$ for the number of believers of the sender's claims,

$$\text{own gain} = \frac{T^{(0)}_{\text{liar}} - T^{(0)}_{\text{silent}}}{b_{\text{liar}}}, \qquad \text{benefit} = \frac{\bar{T}^{(-0)}_{\text{reporter}} - \bar{T}^{(-0)}_{\text{silent}}}{b_{\text{reporter}}},$$

$$\text{harm} = \frac{N\,(\bar{T}_{\text{silent}} - \bar{T}_{\text{all-liars}}) + \text{own gain} \cdot b_{\text{all-liars}}}{(N-1)\,b_{\text{all-liars}}},$$

where (0) is robot 0 and (-0) the mean of the other robots. We measure the harm with all robots lying because a single liar among silent robots barely hurts: traffic spreads out. These per-believer values give Γ in Equation (3), and from it the fine threshold φ_{lo} in Equation (11) with $\bar{s} \approx 0$.

Table 4. Notation.

Symbol	Meaning
Environment	
N, K, T	number of agents, arms and rounds
v_k, F, $\bar{v}$	value of arm k, its distribution, and the mean value (unit of credit)
$\mathcal{K}_i^t$, b_i^t	arms known to agent i and the value of its best known arm
c	cost of exploring an unknown arm
u_i^t, π_i	own payoff of agent i and its policy
y_i^t, ϵ_i^t, σ	noisy observation $y_i^t = v_k e^{\epsilon_i^t}$, its noise, and the noise level
θ_i^t	true value of the arm that i reports on
Claims and reliance	
m_i^t, $\varnothing$	claim of sender i and silence
κ_i^t	inflation factor m_i^t / y_i^t (1 means truthful)
d_j^t, ρ_j, h_j^t	whom receiver j follows, its receiver rule, and its history
L	number of rounds a follower stays committed
g_{ji}^t	gain of follower j from i's claim, $L(\theta_i^t - b_j^t)^+$
$\mathcal{F}_i^t$, f_i^t	follow set $\{j : d_j^t = i\}$ and the indicator that it is non-empty
TVC	
τ	tolerance for verification
V_i^t	verification gate, $\mathbb{1}[\lvert m_i^t/\theta_i^t - 1\rvert \le \tau]$
$\bar{B}$	cap on gain credit per follower (in units of $\bar{v}$)
η, w	bonus weight and score width
$S(m, \theta)$, $\bar{s}$	score of a claim and the best score of any claim outside the tolerance
a_i^t, p	random audit and its probability
χ_i^t	claim is checked (audited or followed)
φ, φ_{lo}	fine for an audited false claim and the smallest fine that keeps truth dominant
C_i^t	TVC credit to sender i
Outside gain	
λ	prosocial weight on the others' mean payoff in the reward (0 selfish, 1 fully cooperative)
r_i^t, $\bar{u}_{-i}^t$	total reward of sender i and the mean payoff of the other agents
G_1, D_1	liar's gain and damage to the others from one lie
ξ	cost of posting a claim
Γ	outside gain of a lie over a truthful claim, $(1-\lambda)G_1 - \lambda D_1$
Δ	margin by which truth beats a lie, $p(\eta + \varphi) - \eta\bar{s}$

Table 5. Multi-agent bandit settings.

Setting	Value
Agents N, arms K, rounds T	10, 50, 100
Episodes, independent runs	1000, 32
Exploration cost c	$0.5\bar{v}$
Commitment L	3 rounds
REINFORCE learning rate, discount	0.1, 0.9
Baseline step size	0.05
Calibrated receivers' learning rate	0.3
Popularity credit	$2\bar{v}$ per follower
Accuracy credit	$1\times$ follower's signed gain
Keeping-exploring credit	$3\bar{v}$ per exploration
TVC: τ, $\bar{B}$, η, w, p, φ	0.05, 8, 2, 0.05, 0.2, 8

D.3. Seeds, Statistics and Compute

In the bandit, the 32 runs are independent policies and arm values simulated as one batch from one random generator (seed 0). The confidence interval is $1.96\, s/\sqrt{32}$, where s is the standard deviation of the per-run mean over the last 100 episodes. In the congestion setting, each seed trains from scratch, the metric of a seed is its mean over the last 50 of 150 episodes, and the confidence interval is the t-interval over seeds. All runs use the CPU only (an Apple M4 Max with 14 cores and 36 GB of memory), with seeds and variants distributed over a process pool.

Table 6. MAPF fake-jam settings.

Setting	Value
Robots N, observation radius, steps	12, 3, 256
Episodes, seeds	150, 10
Jam radius, jam queue	2 steps, 3 robots
Avoidance duration, claim lifetime	20, 20 steps
Sceptical receivers' learning rate	0.3
Cost of posting a claim ξ	0.05
Prosocial weight λ	0 (main), swept in $\{0, \dots, 1\}$
REINFORCE learning rate	0.1
Baseline step size, advantage clip	0.05, $[-10, 10]$
Popularity credit	1 per believer
Accuracy credit	$1\times$ steps saved
TVC: τ, $\bar{B}$, η, p, φ	0.05, 8, 2, 0.2, 15

E. Ablation Studies